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MoveOD: A Data Pipeline for Synthesizing Origin-Destination Demand Data Using U.S. Census Information

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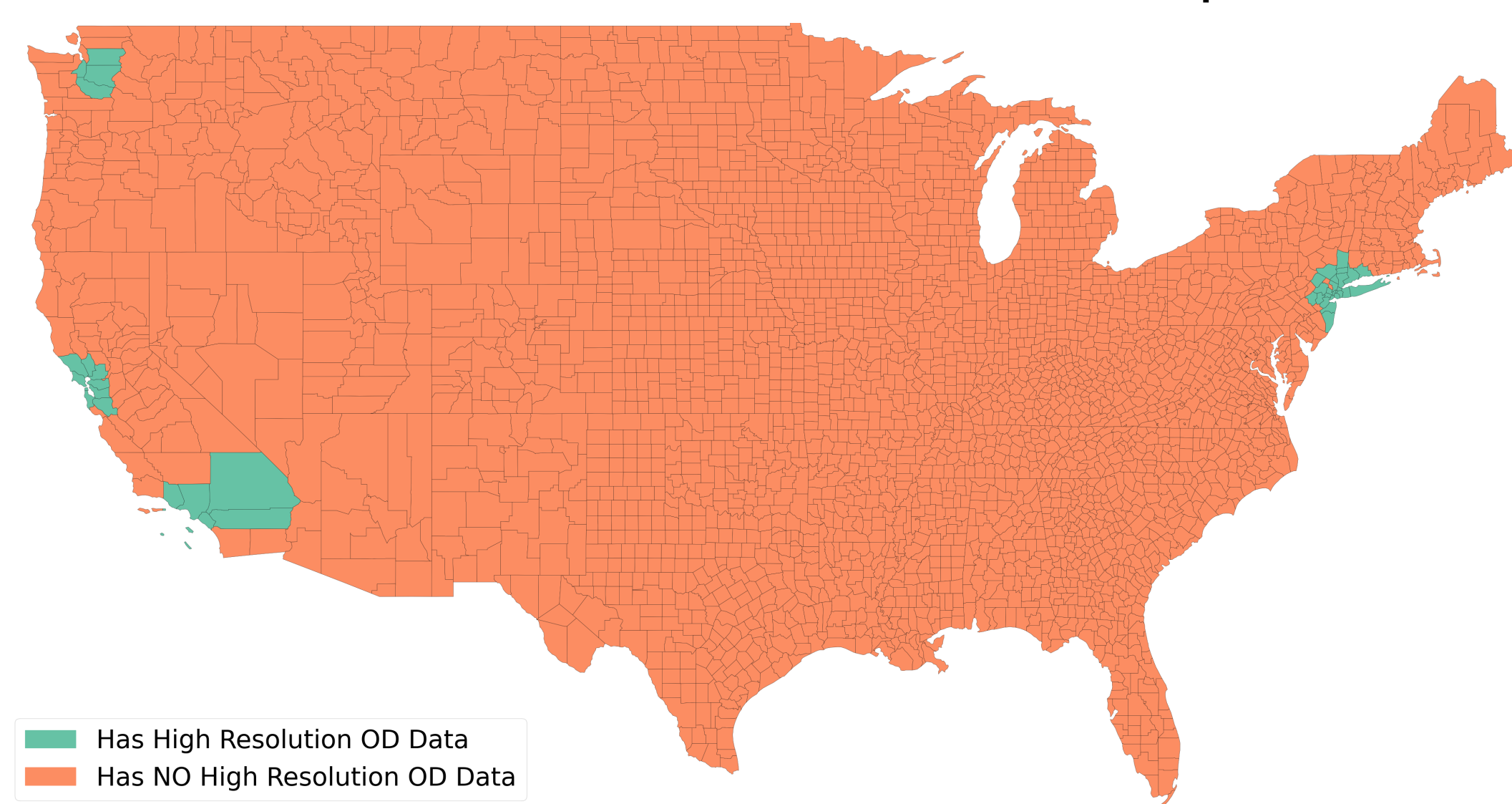


Github link

Motivation

High-resolution Origin-Destination (OD) data is critical for transportation planning, traffic forecasting, and transit optimization. Yet in the United States, such detailed data remains scarce, except a few cities.

- Accurate OD data critical for traffic modeling, routing, and congestion management.
- Traditional methods (surveys, GPS traces) are insufficient due to sparseness, cost, and bias.
- Roadside sensor data is localized and expensive.



Problem description

The goal of MoveOD is to match public census marginals at multiple levels:

- Total commuters per origin region
- Destination shares from LODES data
- Departure-time block distributions from Census Table B08302
- Travel-time distributions from Census Table B08303

Main Assumption - Independence of distributions:

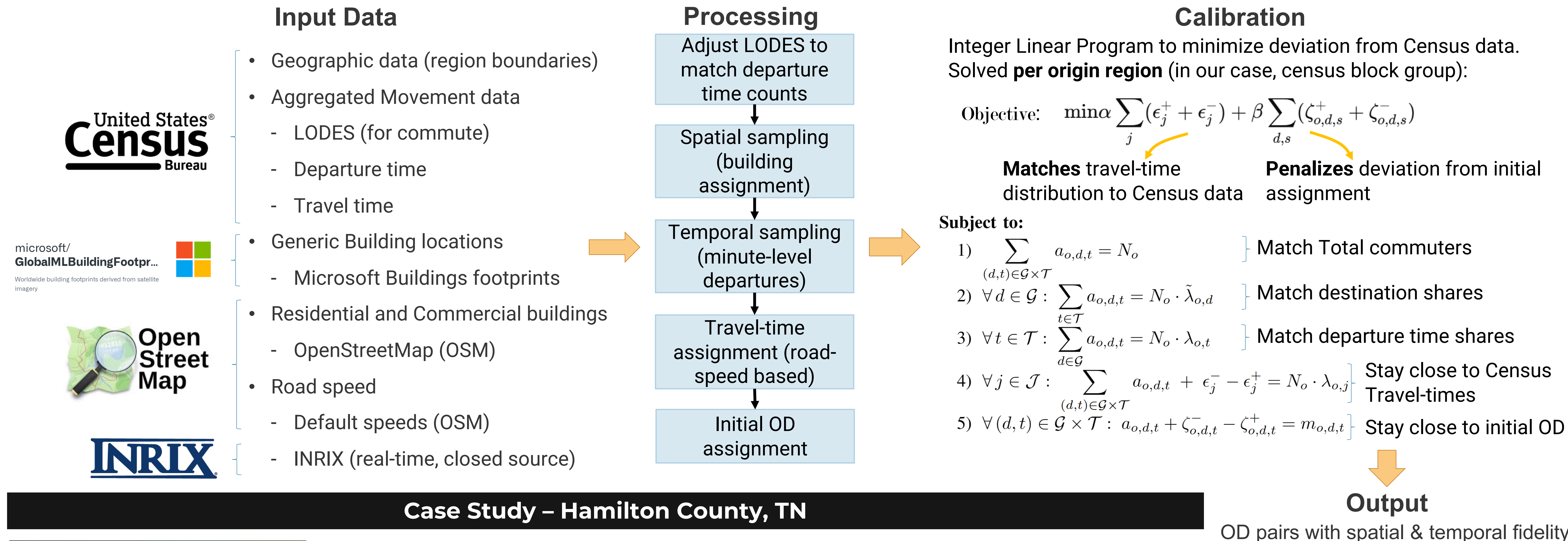
Initial OD assignments assume independence between origin-destination and origin-departure time distributions. This simplifying assumption is explicitly used during the final calibration step to ensure consistency with observed joint patterns.

Features

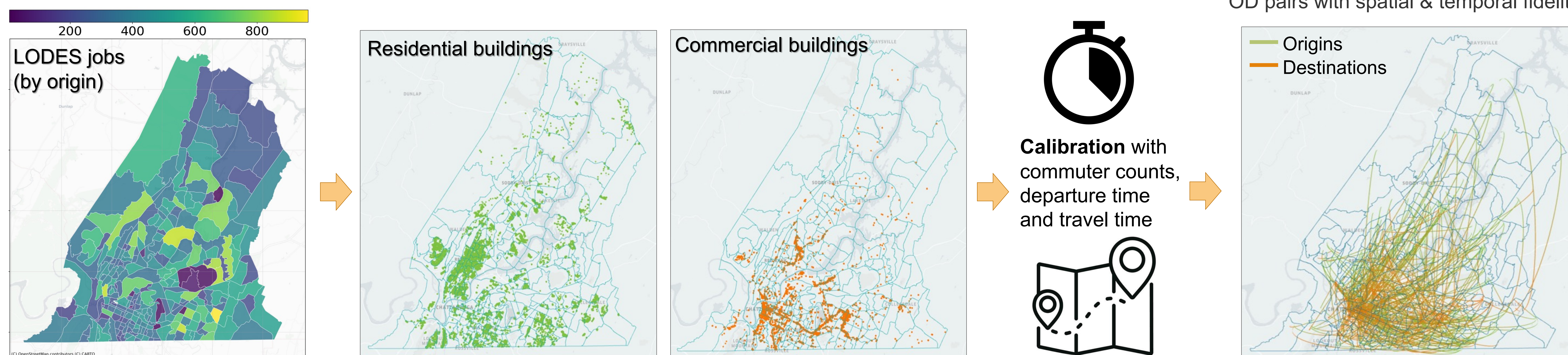
- **Nationwide Scalability:** Applicable to any U.S. county (using Census data) and other countries with available Census data.
- **Temporal-Spatial Granularity:** Minute-level departure times, building-level spatial resolution.
- **Open-Source & Transparent:** Fully reproducible and adaptable pipeline.
- **Statistical Realism:** Synthesized OD data matches multiple Census data marginals (Census, LODES).

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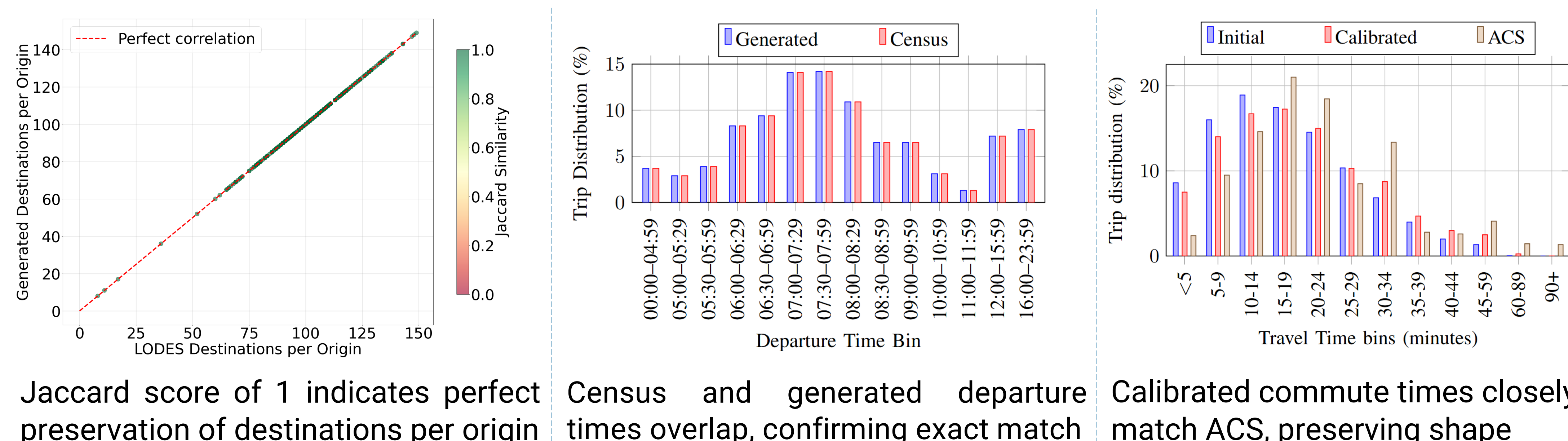
Data Sources & Pipeline for Point-to-Point OD Generation



Case Study – Hamilton County, TN



Results & Validation



Benchmarking

Using Hamilton County OD data using **30 vehicles** (capacity 4) and **100 trips** to perform capacitated routing while minimizing vehicle miles travelled.

Algorithm	VMT ↓	VMT/PMT ↓	Empty ↓ (%)	Description
LKH-3 ^[1]	2,255	1.55	41	SOTA exact/heuristic hybrid.
Clarke-Wright ^[2]	2,623	1.80	50	Classical heuristic.
POMO ^[3]	3,297	2.26	56	Deep RL with policy optimization.

VMT = Vehicle Miles Travelled; VMT/PMT = efficiency ratio; Empty = % distance empty; Coverage = % passengers served.

[1] Helsgaun, K. (2017). "An Extension of the Lin-Kernighan-Helsgaun TSP Solver for Solving the Generalized Traveling Salesman Problem." *European Journal of Operational Research*
 [2] Clarke, G., & Wright, J. W. (1964). "Scheduling of Vehicles from a Central Depot to a Number of Delivery Points." *Operations Research*
 [3] Kwon, Y., Kim, J., Kim, J., & Bengio, Y. (2020). "POMO: Policy Optimization with Multiple Optima for Reinforcement Learning." *NeurIPS*